

EE6402

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 1 EXAMINATION 2025-2026****EE6402 – REAL-TIME DSP DESIGN AND APPLICATIONS**

November / December 2025

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a restricted open book examination. One double-sided A4-size reference sheet with texts handwritten or typed on the A4 paper without any attachments (e.g., sticky notes, post-it notes, gluing or stapling of additional papers) is allowed in the examination hall.
5. Unless specifically stated, all symbols have their usual meanings.

1. (a) (i) Express $a = (1010.110)_2$'s, a two's complement number, as a decimal number.
 (ii) Express $b = (3)_{10}$ in the same format as a . Then add a and b in binary, and express the sum in the same format as a . Is there any overflow?
 (iii) If b is subtracted from a in the same format, will there be any overflow?
 (iv) Express a as a fixed point number in Q6.5 format.
 (10 Marks)
- (b) An analog signal $x_c(t)$ takes the values from -1 v to 1 v, and $x_c(t)$ has a probability density function (pdf) with 0 mean and variance of $\frac{1}{6}$. $x_c(t)$ must be digitized by an n -bit quantizer, such that the SQNR (signal to quantization noise ratio) is at least 23 dB. Determine the minimum possible (integer) value of n .
 (5 Marks)

Note: Question No. 1 continues on page 2.

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- (c) The input to a cascaded system (Figure 1) contains input noise $e_x(n)$ due to quantization of the input with quantization step size Q . First, the input is filtered through a low pass filter with frequency response $|H(e^{j\omega})| = \begin{cases} 1 & 0 \leq \omega \leq 3\pi/4 \\ 0 & 3\pi/4 < \omega \leq \pi \end{cases}$, resulting in the first output noise $e_{total}^{(1)}(n)$. The first filter has no multiplier quantization error and no output quantization error. Second, the first output is filtered through a high pass filter with frequency response $|G(e^{j\omega})| = \begin{cases} 0 & 0 \leq \omega < \pi/2 \\ 1 & \pi/2 \leq \omega \leq \pi \end{cases}$, resulting in the second output noise $e_{total}^{(2)}(n)$. The second filter also has no multiplier quantization error and no output quantization error.

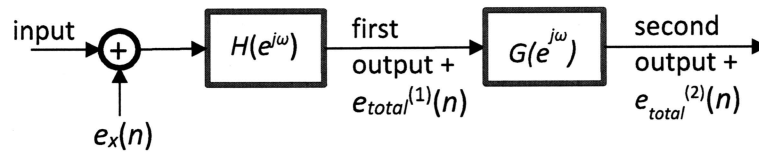


Figure 1

- (i) If truncation is used for input quantization, find the mean $\eta_{total}^{(1)}$ and the variance $(\sigma_{total}^{(1)})^2$ of the first output noise $e_{total}^{(1)}(n)$.
- (ii) If truncation is used for input quantization, find the mean $\eta_{total}^{(2)}$ and the variance $(\sigma_{total}^{(2)})^2$ of the second output noise $e_{total}^{(2)}(n)$.
- (iii) If rounding is used for input quantization, find and sketch the power spectral densities $S_{total}^{(1)}(e^{j\omega})$ and $S_{total}^{(2)}(e^{j\omega})$ of the first and second output noises $e_{total}^{(1)}(n)$ and $e_{total}^{(2)}(n)$.

(10 Marks)

2. (a) A band-pass signal of bandwidth 40 kHz is centered around 120 kHz. Find all possible sampling frequencies for the signal such that there is no aliasing.

(5 Marks)

- (b) A sigma-delta ADC (analog to digital converter) is used to digitize an analog signal, sampled at its Nyquist rate f_s , producing a zero-mean digital signal with the power spectral density shown in Figure 2 on page 3. The quantization error is found to have a variance $\sigma_e^2 = 15$.

Note: Question No. 2 continues on page 3.

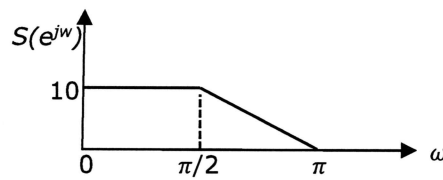


Figure 2

- (i) Find the signal variance σ_x^2 . Hence, find the SQNR of this ADC.
- (ii) If the sampling frequency is increased to $4f_s$, explain how oversampling reduces the quantization error in this sigma-delta ADC. Draw the power spectral density of the oversampled digital signal, and the power spectral density of the quantization error (assume it is white).
- (iii) How much improvement in SQNR is obtained when the sampling is increased from f_s to $4f_s$? (Assume no noise shaping.)

(10 Marks)

- (c) A full adder is shown in Figure 3. All 2-input gates have a delay of 3, while all 3-input gates have a delay of 5.

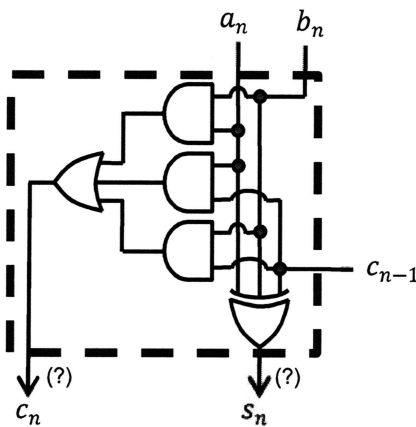


Figure 3

- (i) Find the delays to compute the adder outputs, sum s_n and carry c_n .
- (ii) Draw a diagram of a 3-bit ripple carry adder using the full adder, and show the delays to compute each individual sum and carry outputs.
- (iii) Find a formula for the maximum delay of a N -bit ripple carry adder using this full adder. (Here, maximum means maximum of all N sum and all N carry delays.)

(10 Marks)

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3. (a) Briefly describe how an N -point FFT (fast Fourier transform) achieves the algorithm strength reduction, with emphasis on the operations required to perform the FFT. (5 Marks)
- (b) Implement the FIR filter

$$y(n) = 4x(n) + 6x(n-1) + 10x(n-2) + 15x(n-3):$$
 (i) Draw the conventional implementation using 4 multipliers and 3 adders.
 (ii) Design an implementation using 4 multipliers and 2 adders. Discuss whether this implementation has any disadvantage compared to the conventional approach. (10 Marks)
- (c) Implement the recursive filter $y(n) = 4x(n) + 2y(n-1) - 2y(n-2)$ using 2-latch multipliers and non-pipelined (no latch) adders. Draw the complete structure and label all multiplier values. For an input $x(n)$, if the output of your pipelined implementation is $y(*)$, determine the value of $*$. (10 Marks)
4. (a) The sequential FIR filter shown in Figure 4 is implemented with multipliers having a delay of 26 ns and an adder having a delay of 5 ns. The sampling frequency of $x(n)$ is 50 MHz.

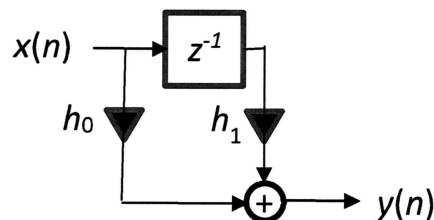


Figure 4

- (i) Draw the data flow graph of Figure 4. The data flow graph should have 4 nodes corresponding to $x(n)$, multiplier (h_0), multiplier (h_1), and adder.
- (ii) Find if the sequential implementation is sufficient for 50 MHz sampling.
- (iii) Use unfolding to obtain a parallel implementation of the filter in Figure 4. Choose the smallest unfolding factor J that is required for the given sampling frequency. Draw the unfolded parallel architecture. (10 Marks)

Note: Question No. 4 continues on page 5.

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(b) TMS320C5515 has four accumulators AC0-AC3.

- (i) What is the length, in bits, of each accumulator?
- (ii) Instead of accessing all the bits simultaneously, certain subsets of the bits may be accessed individually. What are these subsets called, and which bits belong to each subset?

(6 Marks)

(c) Some auxiliary registers and memory addresses in a TMS320C55x contain the following values:

XAR0	00_0100h
XAR1	00_0200h
XAR2	00_0300h
00_0100	0300h
00_0101	0301h
00_0102	0001h
00_0200	0100h
00_0201	0102h
00_0202	0002h
00_0300	0200h
00_0301	0003h

Using the operands rules for AR indirect addressing, find the values of **XAR0**, **XAR1** and **XAR2** after each of the following consecutive instructions:

```
MOV *AR0, AR1
MOV *AR1+, AR2
MOV ++AR2, AR0
```

(9 Marks)

END OF PAPER

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EE6402 REAL-TIME DSP DESIGN & APPLICATIONS

Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.